

Paper 1

No.: 1

Title: Energy-Efficient Clustered Wireless Sensor Networks for Border Surveillance

Authors: Malathy et al.

Year: 2024

Thought (comparison):

Malathy and colleagues investigate energy-efficient clustered wireless sensor network (WSN) architectures for border surveillance, integrating ant colony optimization (ACO) algorithms for routing and ECC cryptography for secure long-duration operation. Their emphasis on hierarchical cluster management and energy-aware routing addresses lifetime extension in static, infrastructure-light deployments. In contrast, our project focuses on a single-node expansion board for ESP32 devices that improves local power stability, real-time sensing, and cloud ML — a device-level hardware/firmware solution rather than a network-level routing optimization. While their energy-efficiency goals are conceptually relevant (both aim to maximize uptime), the methods (ACO clustering vs. local power conditioning, heatsinks, and MQTT telemetry) and application assumptions differ substantially.

Match Factor: 35%

Paper 2

No.: 2

Title: Integrated Architecture for Border Surveillance with Heterogeneous Sensor Fusion

Authors: Essendorfer et al.

Year: 2024

Thought (comparison):

Essendorfer et al. propose an end-to-end architecture that fuses heterogeneous sensors (radar, camera, thermal, acoustic) to enhance detection accuracy for border surveillance, focusing on middleware and data-level fusion strategies. Their work provides useful insight into multi-sensor data aggregation and latency/throughput trade-offs for real-time systems. However, their emphasis is on system-level sensor fusion algorithms and heterogeneous network coordination rather than microcontroller power stabilization or Wi-Fi-induced brownouts. For our ESP32 expansion board, the paper's sensor fusion concepts could inspire how to pre-process and label telemetry before sending to cloud ML, but it does not address the core hardware reliability and decoupling issues we solve.

Match Factor: 25%

Paper 3

No.: 3

Title: Swarming Control Algorithms for Unmanned Ground and Aerial Vehicles in Surveillance and Infrastructure Protection

Authors: Sauter et al.

Year: 2024

Thought (comparison):

Sauter and coauthors analyze control algorithms for coordinated swarms of UGVs and UAVs, with mathematical treatment of formation keeping, coverage, and resilience to failures. The contribution is primarily in control theory and multi-agent coordination, valuable when designing high-level behaviors for autonomous units. Our project is hardware-centric — ensuring stable power, thermal reliability, and robust MQTT telemetry on ESP32 units — so swarm control algorithms are orthogonal

to our contributions. The paper might inform future use-cases where multiple stabilized ESP32 units operate cooperatively, but it does not inform capacitor selection, brownout prevention, or ESP32 firmware strategies.

Match Factor: 15%

Paper 4

No.: 4

Title: IoT-based Smart Border Security System Utilizing Machine Learning for Intruder Detection

Authors: Fatima et al.

Year: 2024

Thought (comparison):

Fatima et al. describe an IoT pipeline combining edge sensing, cloud ML classification, and alerting for intruder detection — an architecture similar in concept to our MQTT → PC → cloud ML pipeline. Their focus on feature extraction and classifier models for detection is directly relevant to the ML portion of our work. However, their treatment likely assumes reliable edge devices and does not dig into microcontroller power-conditioning or brownout mitigation. Thus, the paper maps well to our analytics and model selection needs but offers limited guidance on the hardware stabilization features (bulk capacitors, heatsinks, voltage-display UX) that are central to our expansion board.

Match Factor: 40%

Paper 5

No.: 5

Title: Wireless Sensor Network Techniques for Border Surveillance and Intruder Detection: A Survey

Authors: Arjun et al.

Year: 2024

Thought (comparison):

Arjun and coauthors survey WSN methods for surveillance, covering deployment strategies, routing protocols, and energy-saving mechanisms — useful for system architects planning large sensor deployments. For our single-node ESP32 expansion board, the survey provides background on the trade-offs between centralized vs. decentralized sensing and on minimizing energy consumption at network scale. Still, the paper is network-centric and mostly simulation/architecture-focused, whereas our contribution is practical hardware reliability under Wi-Fi/Bluetooth loads and ensuring continuous sensor telemetry to ML systems.

Match Factor: 30%

Paper 6

No.: 6

Title: Machine Learning Approaches: A Comprehensive Survey

Authors: Thomas et al.

Year: 2024

Thought (comparison):

Thomas et al. present a broad compendium of ML techniques, model architectures, and evaluation metrics — a useful reference when choosing anomaly-detection or regression models for telemetry prediction. Their survey can guide selection among classical and deep methods and suggest best

practices for model validation, which we can apply to cloud analytics for temperature/current anomaly detection. However, it is generic across domains and does not address IoT-specific constraints like on-device inference on an ESP32, nor the effects of intermittent connectivity due to brownouts. Thus its utility is greatest for ML design rather than hardware design.

Match Factor: 50%

Paper 7

No.: 7

Title: YOLO-based Real-Time Border Security Surveillance System

Authors: Nahata et al.

Year: 2024

Thought (comparison):

Nahata et al. implement a YOLO object-detection pipeline tailored for real-time surveillance cameras, focusing on model optimization and deployment on constrained compute (likely edge GPUs or embedded SoCs). The relevance to our project lies in the potential use of ESP32-CAM modules for vision tasks; however, YOLO pipelines commonly stress network and compute resources and assume different hardware capabilities than the ESP32 microcontroller. Their work helps understand heavy Wi-Fi/video streaming loads that can destabilize an ESP32 — indirectly motivating our power-stability improvements — but does not address microcontroller power-conditioning directly.

Match Factor: 15%

Paper 8

No.: 8

Title: Autonomous Wireless Sensor Network Employing Swarm of Mobile Robots for Border Intrusion Detection

Authors: Mahjoub et al.

Year: 2024

Thought (comparison):

Mahjoub et al. propose distributed mobile sensing with cooperative decision-making across a robot swarm, emphasizing autonomy and network resilience. The distributed, mobile nature parallels our interest in autonomous ESP32-CAM nodes, but their focus is on multi-robot collaboration, path planning, and mission-level robustness rather than device-level power integrity. For scenarios where many ESP32 units operate continuously and wirelessly, their system-level insights are useful; nevertheless, the paper won't provide practical guidelines on capacitor selection or thermal design for an ESP32 expansion board.

Match Factor: 20%

Paper 9

No.: 9

Title: Mini Seeker Robot: Automated Border Surveillance Platform with Facial Recognition

Authors: Reddy et al.

Year: 2024

Thought (comparison):

Reddy et al. build a robotics platform integrating cameras and facial recognition; their work

concentrates on vision pipelines, onboard compute, and remote streaming. The hardware platform and continuous camera streaming use-cases are conceptually relevant because they highlight Wi-Fi and compute loads that can cause brownouts on ESP32 devices. However, the paper's primary contribution (facial recognition system and robot integration) diverges from our power-stability and telemetry-focused expansion board; it does not discuss power conditioning or capacitor networks needed to avoid resets under heavy wireless loads.

Match Factor: 10%

Paper 10

No.: 10

Title: Helmet Detection and Number Plate Recognition using YOLOv5 and EasyOCR

Authors: Sathe et al.

Year: 2024

Thought (comparison):

Sathe et al. apply YOLOv5 and EasyOCR for safety/traffic recognition tasks — heavy computer-vision and OCR workloads that are irrelevant to microcontroller-level power stability. Their methods are valuable if an ESP32-CAM is used merely as a camera upstream to a stronger edge device, but the paper doesn't inform on ESP32-specific firmware, decoupling capacitors, or MQTT telemetry reliability. The work is therefore peripheral and more applicable to model selection for vision tasks than to our hardware design concerns.

Match Factor: 5%

Paper 11

No.: 11

Title: Digital Border Surveillance System (DBSS) using YOLOv7 for Real-Time Object Detection

Authors: Noor et al.

Year: 2024

Thought (comparison):

Noor et al. present a YOLOv7-based detection pipeline for surveillance that emphasizes detection accuracy and system latency. While DBSS demonstrates the end-to-end demands of real-time video processing, it assumes robust compute and networking infrastructure unlike our constrained ESP32 context. The paper signals the high data rates and streaming scenarios that can destabilize embedded Wi-Fi devices, reinforcing the need for our board's improved power delivery; however, it lacks concrete guidance on microcontroller-level power mitigation strategies.

Match Factor: 10%

Paper 12

No.: 12

Title: Deep Learning Framework using YOLOv5 Variants for Pedestrian Detection in Thermal Images

Authors: Kumar et al.

Year: 2024

Thought (comparison):

Kumar et al. tackle thermal-image pedestrian detection using YOLO variants, including dataset curation and model optimization for thermal sensors. This is relevant only insofar as thermal sensors

may be part of a multi-sensor ESP32 node; otherwise, the core focus is on CV model performance and not on ESP32 hardware or MQTT telemetry. It offers little direction on preventing brownouts or designing decoupling networks, though thermal-sensor integration could map to our telemetry pipeline.

Match Factor: 10%

Paper 13

No.: 13

Title: YOLO-based Real-Time Surveillance System for Persons, Vehicles, Weapons, and Drones Detection

Authors: Taneja et al.

Year: 2024

Thought (comparison):

Taneja et al. propose a multi-class YOLO-based detection system covering persons, vehicles, weapons and drones — a CV-heavy application with high bandwidth requirements. The paper is valuable to understand worst-case loads on wireless links (helpful for stress-testing our ESP32 expansion board), but it does not contribute to the hardware-level power conditioning or MCU stability improvements that are central to our design.

Match Factor: 10%

Paper 14

No.: 14

Title: IoT-based MPPT Solar Charger with Arduino UNO R4 WiFi and WebSocket Communication

Authors: Daou et al.

Year: 2024

Thought (comparison):

Daou and colleagues present a practical MPPT solar charger integrated with an Arduino UNO R4 WiFi module, including WebSocket comms — directly relevant to power-management strategies and robust IoT communications. Their hardware-oriented treatment of power electronics and energy harvesting provides applicable ideas for managing input power stability and designing protective/regulation circuits, although the MCU platform differs from an ESP32. Concepts such as smoothing transients, input regulation, and reliable comm protocols can inform our expansion board's upstream power-handling and telemetry reliability.

Match Factor: 55%

Paper 15

No.: 15

Title: 60 GHz CMOS-SOI Transceiver for Internet of Things Nodes

Authors: Saponara et al.

Year: 2024

Thought (comparison):

Saponara et al. detail a high-frequency CMOS-SOI transceiver design for IoT nodes, focusing on RF front-end performance and integration. While informative for RF and antenna considerations, the paper's focus on mmWave hardware and SoC integration is tangential to the ESP32's 2.4 GHz Wi-

Fi/Bluetooth operation and to microcontroller-level power conditioning. Unless our expansion board needs custom RF front-ends or interference mitigation at mmWave bands, the relevance is limited.
Match Factor: 35%

Paper 16

No.: 16

Title: IoT-powered Smart Surveillance Sentinel Bot with Mobile Robot Platform

Authors: Geetha M et al.

Year: 2024

Thought (comparison):

Geetha et al. present an integrated robotic sentinel using IoT sensors and remote monitoring — similar in system architecture to multi-node deployments that might include our ESP32 expansion board. The paper's system-level integration lessons (sensor interfacing, telemetry handling, remote dashboards) are applicable to our pipeline design. However, it does not address device-level thermal or power conditioning techniques, which are our unique contribution. Thus, it's useful for high-level integration patterns rather than for hardware stabilization best practices.

Match Factor: 20%

Paper 17

No.: 17

Title: Navigation Interface for Multiple Autonomous Mobile Robots

Authors: Sergio Saponara et al.

Year: 2024

Thought (comparison):

Saponara et al. focus on navigation interfaces for coordinated robots, touching on UI, path planning, and control loops. This is system-level robotics work that may use telemetry streams similar to ours but does not tackle ESP32 power integrity or decoupling strategies. It could be relevant if our expansion board will be used in mobile robotics where stable power is critical, but the direct technical overlap is small.

Match Factor: 15%

Paper 18

No.: 18

Title: AI-based Desktop Program for Human Detection using OpenCV and Haar-cascade Classifiers

Authors: Alajrami et al.

Year: 2024

Thought (comparison):

Alajrami et al. implement a desktop computer-vision pipeline (OpenCV + Haar cascades) for human detection — a classic but resource-heavy approach performed on PC-class hardware. This is only peripherally relevant: if our ESP32 nodes offload vision to a PC, the paper's detection pipeline could be replicated on the PC side of our MQTT → PC → cloud flow. It does not help with microcontroller power or ESP32 firmware stability.

Match Factor: 10%

Paper 19

No.: 19

Title: AI-based Human Identification for Selective Recording in Surveillance Systems

Authors: Eman Alajrami et al.

Year: 2024

Thought (comparison):

This work explains selective recording triggered by AI detections to reduce bandwidth and storage. From our perspective, selective transmission strategies are relevant — sending only important telemetry or images reduces Wi-Fi duty and can mitigate power spikes caused by continuous streaming. But the paper addresses high-level policy and ML thresholds rather than hardware solutions to brownouts, so it complements our work on the data-flow side rather than the hardware side.

Match Factor: 10%

Paper 20

No.: 20

Title: Next-Generation Defense Monitoring System with Digital Twin, Blockchain, and AI-based Facial Recognition

Authors: Geetha et al.

Year: 2024

Thought (comparison):

Geetha et al. combine digital twin models, blockchain audit trails, and AI for facial recognition — an ambitious system-level architecture that emphasizes reliability, auditability, and simulation. For our project, the “digital twin” and telemetry verification concepts are useful for validation and anomaly detection workflows, but blockchain and facial-recognition layers are not central to ESP32 power stability. The paper’s systems perspective may inspire end-to-end testing frameworks for our expansion board.

Match Factor: 20%

Paper 21

No.: 21

Title: Blockchain for Decentralization and Auditability in Defense Systems

Authors: Mohith S et al.

Year: 2024

Thought (comparison):

This paper explores blockchain for tamper-evident logging and decentralized coordination; such mechanisms might be useful in high-assurance deployments of ESP32 telemetry where audit trails are required. Nonetheless, the direct relevance to microcontroller-level power conditioning, decoupling, or thermal management is minimal. The main utility is conceptual: how to store/verify telemetry and ML outputs securely in distributed systems.

Match Factor: 15%

Paper 22

No.: 22

Title: Privacy-preserving and Efficient Border Surveillance System

Authors: Ramyashree P M et al.

Year: 2024

Thought (comparison):

Ramyashree et al. address privacy techniques (e.g., selective anonymization, data minimization) for surveillance systems, which aligns with responsible telemetry collection in IoT. For our project, privacy-preserving strategies can inform what data we transmit from ESP32 nodes and how cloud ML handles sensitive signals. However, these concerns are orthogonal to the hardware-level stabilization techniques (capacitors, heatsinks, voltage displays) we implement.

Match Factor: 20%

Paper 23

No.: 23

Title: Human Detection Rover for AI-Powered Search and Rescue in Disaster Zones

Authors: Dilbar Roshan et al.

Year: 2024

Thought (comparison):

Dilbar Roshan et al. present an AI-enabled rover for search & rescue, integrating multiple sensors and wireless telemetry. The platform-level insights into robustness, sensor sampling strategies, and reliability in harsh environments are useful analogs for deploying ESP32 nodes in demanding conditions. Yet, the paper focuses on robotics and mission-level resilience rather than explicit microcontroller power-conditioning practices, limiting direct applicability to our expansion board's electrical design.

Match Factor: 15%

Paper 24

No.: 24

Title: Hybrid Bot-Detection Framework Combining Reinforcement Learning and Swarm Optimization for Social Networks

Authors: Greeshma Lingam et al.

Year: 2024

Thought (comparison):

This is a cyber/social-network security paper applying RL and swarm optimization to detect bots online — conceptually interesting but unrelated to embedded IoT hardware, power integrity, or MQTT telemetry. It offers no guidance on ESP32 stability or thermal design.

Match Factor: 5%

Paper 25

No.: 25

Title: Portable GPS Tracking Solution for Real-Time Outdoor Monitoring

Authors: Sudhanshu Janwadkar et al.

Year: 2024

Thought (comparison):

Janwadkar et al. detail a compact GPS tracking system with real-time telemetry and power-optimized

operation, which aligns partially with IoT telemetry and low-power design concerns. Their experience with intermittent connectivity and power budgeting can inform strategies for continuous sensing while prolonging uptime — relevant when designing power-hungry features like Wi-Fi streaming on ESP32. Still, the paper probably focuses more on GPS and comms than on fine-grained MCU decoupling and heatsink design.

Match Factor: 30%

Paper 26

No.: 26

Title: FPGA-based Rescue Bot for Surveillance with Internet-Controlled Mobility and Sensor Fusion

Authors: K. Sudhakar et al.

Year: 2022

Thought (comparison):

Sudhakar et al. deploy FPGA-based platforms for robotic rescue with sensor fusion; FPGAs offer performance and I/O advantages but are a different hardware paradigm than ESP32 microcontrollers. The paper's sensor-fusion and internet-control lessons may translate to architecture-level choices in our dashboard and ML pipeline, yet it does not address ESP32-specific power brownouts or capacitor networks. As such, the relevance to our expansion board design is limited.

Match Factor: 15%

Paper 27

No.: 27

Title: Laser Beam Intrusion Detection Mechanism for Outdoor Perimeter Security

Authors: Anisom Chakraborty et al.

Year: 2024

Thought (comparison):

Chakraborty et al. design a laser-beam-based intrusion detection sensor, which is an example of specialized sensor hardware. Their attention to reliable sensing and false-positive reduction is conceptually relevant for telemetry quality, but laser-beam hardware design and peripheral circuitry differ markedly from the power-conditioning and Wi-Fi stability problems we solve. The paper could inspire candidate sensors for integration but not the core electrical design techniques used on our ESP32 board.

Match Factor: 20%

Paper 28

No.: 28

Title: Building Monitoring System with Multi-Sensor Fusion for False Alarm Reduction

Authors: Tao Qi et al.

Year: 2024

Thought (comparison):

Tao Qi et al. focus on sensor-fusion algorithms and decision logic to reduce false alarms in building monitoring — directly relevant to ML/data-processing aspects of our project. Their methods for combining temperature, current, occupancy, and other signals can be applied to the cloud ML model we use to detect anomalies from ESP32 telemetry. The hardware-level stabilization remains out of

scope for them, but their fusion strategies can meaningfully improve our anomaly detection performance.

Match Factor: 45%

Paper 29

No.: 29

Title: Autonomous Robot with IoT Connectivity and On-Board Sensing for Remote Monitoring

Authors: Muhamad Syazmie Sepeeh et al.

Year: 2024

Thought (comparison):

Sepeeh et al. present an autonomous robot with integrated IoT telemetry and on-board sensors — a system-level implementation that shares telemetry, connectivity, and robustness concerns with our deployment scenario. Their discussion of connectivity strategies and remote monitoring are applicable, but again their hardware choices (robotics-grade controllers) differ from an ESP32 microcontroller. The paper provides useful system-integration perspective rather than microcontroller power-conditioning specifics.

Match Factor: 25%

Paper 30

No.: 30

Title: Compact Surveillance Robot using ESP32-CAM for Live Video Transmission

Authors: Gurpreet Kaur et al.

Year: 2024

Thought (comparison):

Gurpreet Kaur's work is directly relevant: it uses **ESP32-CAM**, documents live video streaming over Wi-Fi, and thus implicitly deals with the practical limitations of ESP32 hardware under heavy network and sensor load. Their implementation likely reports on buffering, dropped frames, and thermal/load behavior — insights directly applicable to our expansion board's motivation (preventing brownout resets and overheating). While their focus may be application-level, their empirical observations about ESP32 performance under Wi-Fi loads make this paper highly useful for firmware and stress-testing approaches.

Match Factor: 60%

Paper 31

No.: 31

Title: IoT-enabled Health Monitoring System using ESP32 and NODEMCU ESP8266 with Multi-Sensor Integration

Authors: Balamaniikandan A et al.

Year: 2024

Thought (comparison):

Balamaniikandan et al. present an IoT health-monitoring platform built on **ESP32 and ESP8266**, addressing sensor integration, MQTT telemetry, and data visualization. This is strongly pertinent to our project because it shares the same hardware family and communication patterns; their lessons on sensor sampling rates, MQTT quality-of-service choices, and power-consumption trade-offs can

directly inform our firmware and middleware design. They likely do not focus on advanced power-conditioning components (bulk decoupling networks or heatsinks), but their practical experience with ESP variants makes their findings highly actionable for our telemetry pipeline.

Match Factor: 70%

Paper 32

No.: 32

Title: Low-Cost Modular Physical Robot Platform for Swarm Exploration

Authors: Rebeca Solis-Ortega et al.

Year: 2024

Thought (comparison):

Solis-Ortega et al. design a modular robot platform aimed at swarm exploration — an emphasis on modularity, sensor plug-and-play, and ruggedness. From our viewpoint, modular hardware ideas (e.g., separate sensor modules, swappable power modules) can inform our expansion board's mechanical and connector design choices. However, the paper centers on robotics platform-level concerns rather than microcontroller-level power stability or analytics pipelines, limiting direct technical overlap.

Match Factor: 20%

Paper 33

No.: 33

Title: Comparison of Physical and Higher-Layer Multi-Connectivity Methods for Swarm-based Patrolling

Authors: Marie-Theres Suer et al.

Year: 2024

Thought (comparison):

Suer et al. compare multi-connectivity approaches (physical-layer redundancy, higher-layer failover) to improve resilience in swarm patrolling. Their analysis of connectivity redundancy and protocol trade-offs is relevant when considering how ESP32 nodes maintain telemetry under unstable links. Choosing the right connectivity strategy (e.g., keepalive intervals, reconnection backoff, mesh vs TCP) can reduce retransmission spikes that stress power supply; thus, the paper is pertinent to communication strategy decisions that complement our hardware-level stabilization.

Match Factor: 35%

Paper 34

No.: 34

Title: Sliding-mode Navigation Law for Unicycle-like Vehicles in Boundary Exploration

Authors: Matveev et al.

Year: 2024

Thought (comparison):

Matveev et al. offer control-law contributions for boundary exploration in ground vehicles, focusing on robust guidance under uncertainty. This control-theory content is not relevant to ESP32 power conditioning or MQTT telemetry beyond possible application in mobile platforms using our hardware. The technical overlap is low: the paper helps if our stabilized ESP32 board is integrated

into vehicles requiring robust navigation, but otherwise remains orthogonal.

Match Factor: 10%

Paper 35

No.: 35

Title: Architecture for Ground Bots and Drone Swarms Coordination in Disaster Response

Authors: Himadri Nath Saha et al.

Year: 2024

Thought (comparison):

Saha et al. describe orchestration of heterogeneous ground and aerial bots for disaster response, with emphasis on coordination, communications, and mission planning. The system-level reliability and robustness concerns align with our interest in dependable telemetry, but the paper focuses on mission orchestration rather than device-level power and thermal stabilization. It provides context for large-scale deployments that could benefit from our stabilized ESP32 nodes, but offers limited hardware design guidance.

Match Factor: 20%

Paper 36

No.: 36

Title: Intelligent Swarm-based Bots for Continuous Area Patrolling and Detection with Multi-Sensor IoT Architecture

Authors: Ajay N. Upadhyaya et al.

Year: 2024

Thought (comparison):

Upadhyaya et al. propose an IoT architecture for continuous patrols using intelligent swarm bots and multi-sensor fusion. The architecture-level descriptions (edge processing, data aggregation, alert pipelines) are relevant to how our telemetry is routed and analyzed, but again their work does not delve into microcontroller-level electrical reliability. Their system insights help with scaling and orchestration considerations for networks of our stabilized ESP32 modules.

Match Factor: 25%

Paper 37

No.: 37

Title: Sliding-mode Navigation for Unicycle-like Vehicles Following Unknown Boundaries at Constant Distance

Authors: Matveev et al.

Year: 2024

Thought (comparison):

This appears to be a related Matveev control-theory piece (duplicate topic). The content is focused on motion control and not on ESP32 hardware or telemetry reliability. While useful in robotics contexts, it offers little for designing capacitor networks or heatsinks for microcontrollers.

Match Factor: 10%

Paper 38

No.: 38

Title: IoT Multi-Sensor Fusion with Edge Analytics for Threat Detection in Swarm Patrolling

Authors: Muthukumar K et al.

Year: 2024

Thought (comparison):

Muthukumar and colleagues present an edge-analytics pipeline for fusing multi-sensor inputs to detect threats — directly applicable to our cloud ML and edge pre-processing goals. Their emphasis on reducing false alarms through edge analytics and using lightweight models at the edge can guide which features we compute on the ESP32 vs send to the cloud, helping to manage network load and indirectly reduce power spikes associated with continuous transmission. The paper complements our hardware stabilization by informing telemetry pre-processing and model partitioning decisions.

Match Factor: 45%

Paper 39

No.: 39

Title: Low-Power mmWave Radar for Occupant Detection in Swarm Patrolling Systems

Authors: Muhammet Emin Yanik et al.

Year: 2024

Thought (comparison):

Yanik et al. develop a low-power mmWave radar for occupant detection; their sensor technology is specialized and useful for high-resolution detection. However, mmWave radar integration has different power and interface requirements than the temperature/current sensors on our ESP32 expansion board. The principal relevance is conceptual — a lesson in integrating low-power sensors into IoT nodes — but it doesn't provide guidance on MCU decoupling, brownout prevention, or ESP32 thermal management.

Match Factor: 20%

Paper 40

No.: 40

Title: Bee-Bots: Decentralized Ledger with Swarm Robots through Blockchain for Continuous Patrolling

Authors: N. Praveen Chandhar et al.

Year: 2024

Thought (comparison):

Chandhar et al. explore blockchain for decentralized coordination and logging among swarm robots. The decentralized logging idea might be interesting for tamper-resistant telemetry storage, but blockchain adds complexity and computational overhead unsuited to low-power ESP32 nodes. The paper is therefore mostly orthogonal to our hardware stabilization aims and more relevant to system accountability and auditing aspects.

Match Factor: 10%

Paper 41

No.: 41

Title: Evolution from Manual Patrols to Autonomous Systems: Advanced Unmanned Ground and Aerial Vehicles for Border Security

Authors: M. Alshukri et al.

Year: 2024

Thought (comparison):

Alshukri et al. provide a broad survey of autonomous patrol systems and the shift from manual to autonomous platforms — useful background for the application domain of our ESP32 modules. Their high-level discussion of autonomy, sensor modalities, and deployment challenges contextualizes the use cases for stabilized microcontroller nodes. However, it does not provide technical guidance on the specific electrical engineering problems (brownout, capacitor layout) we address.

Match Factor: 15%

Paper 42

No.: 42

Title: TALOS: Innovative Mobile Border Surveillance Solution Integrating UGVs, UAVs, and Static Towers

Authors: Tanas et al.

Year: 2024

Thought (comparison):

Tanas et al. present a combined architecture integrating UGVs, UAVs, and static towers for persistent surveillance. While this offers a system-level blueprint for heterogeneous deployments and telemetry flows (relevant to our dashboard and cloud ML), the work does not delve into microcontroller-level power conditioning. Our board could serve as the reliable ESP32 endpoint in such an architecture, but the paper itself is not a source for capacitor or thermal design choices.

Match Factor: 15%

Paper 43

No.: 43

Title: Low-Cost Surveillance Robot for Indian Border Security Force with Arduino-based Platform

Authors: Bagyam et al.

Year: 2024

Thought (comparison):

Bagyam et al. describe a low-cost Arduino-based surveillance robot — relevant as an example of constrained hardware platforms used in field deployments. Lessons on reducing cost while maintaining functionality can inspire design choices, but Arduino vs ESP32 platform differences (Wi-Fi capabilities, brownout behaviors) limit direct applicability. The paper is more useful for system-level budgeting and modular design than for detailed ESP32 power-management solutions.

Match Factor: 20%

Paper 44

No.: 44

Title: Internet-Controlled Raspberry Pi Robot with Video Streaming and Facial Recognition for Border Patrol

Authors: Janani et al.

Year: 2024

Thought (comparison):

Janani et al. implement a Raspberry Pi-based video streaming and recognition robot; the Pi's power profile and thermal characteristics differ strongly from ESP32 MCUs. The paper highlights heavy compute and streaming use-cases that could destabilize lower-power nodes, thus motivating our board's stability features, but it does not provide design prescriptions for MCU-level decoupling or brownout mitigation. It's therefore only tangentially useful.

Match Factor: 10%

Paper 45

No.: 45

Title: IoT-enabled Video Surveillance Robot with Night Vision and Passive Infrared Motion Detection

Authors: William et al.

Year: 2024

Thought (comparison):

William et al. combine video, night-vision, and PIR sensors on a robotic platform and discuss telemetry, streaming, and detection pipelines. The integration challenges (sensor fusion, event-triggered streaming) are relevant for designing telemetry strategies that avoid continuous high-power Wi-Fi usage on ESP32 devices. However, the hardware platforms discussed are likely more powerful than an ESP32, so their electrical-design recommendations are not directly transferable to our capacitor/heatsink work.

Match Factor: 15%

Paper 46

No.: 46

Title: Link Layer Security Architecture for Military IoT Devices in Low-Power Wireless Sensor Networks

Authors: Furtak et al.

Year: 2024

Thought (comparison):

Furtak et al. discuss link-layer security for low-power IoT devices, covering encryption, key management, and secure boot strategies tailored to constrained nodes. Security considerations are important for our telemetry pipeline (secure MQTT, authenticated ML inputs), and their low-power focus overlaps with the need to avoid excessive CPU/network overhead that could lead to power spikes or increased energy draw. While the paper is not about power hardware design, its guidance helps choose secure, low-overhead communication schemes compatible with our stabilized ESP32 nodes.

Match Factor: 40%

Paper 47

No.: 47

Title: Experimental Comparison of ESP-MESH, ESP-NOW, and TCP/IP for Remote Monitoring on ESP8266-based Systems

Authors: Aznabaev et al.

Year: 2024

Thought (comparison):

Aznabaev et al. directly compare communication protocols (ESP-MESH, ESP-NOW, TCP/IP) on Espressif hardware, providing empirical results on latency, throughput, and reliability. This paper is highly relevant because protocol choice and network behavior can directly influence Wi-Fi activity bursts and thus power draw and brownout potential on ESP32 devices. Although focused on ESP8266, the findings map well to ESP32 and can guide firmware-level choices (e.g., avoid continuous TCP streaming, prefer low-overhead ESP-NOW for specific telemetry) that complement our hardware stabilization. This makes it one of the most actionable references for our work.

Match Factor: 85%

Paper 48

No.: 48

Title: Swarm of Vibration-driven Swarm-bots for Exploration of Collective Motion Behaviors and Jamming Transitions

Authors: Rozenblit et al.

Year: 2024

Thought (comparison):

Rozenblit et al. study physical swarm dynamics of vibration-driven bots, focusing on emergent collective behaviors and jamming phenomena. This research is primarily in embodied robotics and emergent dynamics — interesting for swarm behavior modeling but irrelevant to MCU-level electrical design, MQTT telemetry, or cloud ML integration. It provides little practical guidance for ESP32 hardware stabilization.

Match Factor: 10%

Paper 49

No.: 49

Title: Advanced Border Intrusion Detection and Surveillance using Wireless Sensor Network Technology

Authors: H. A. Omar & T. N. Saadawi

Year: 2013

Thought (comparison):

Omar & Saadawi present an earlier, foundational treatment of WSN-based intrusion detection, outlining architectures, sensor types, and trade-offs in networked monitoring. While somewhat dated, the paper offers useful background on sensor-network principles and resource-constrained design that can inform high-level architectural decisions. Nevertheless, it precedes modern ESP32-era Wi-Fi challenges and does not address modern brownout mitigation, so its direct technical applicability to our board's electrical design is limited.

Match Factor: 25%

Paper 50

No.: 50

Title: Path Planning Method Considering Blind Spots based on ROS Navigation Stack and Dynamic Window Approach for Wheeled Mobile Robot

Authors: M. Kobayashi & N. Motoi

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Thought (comparison):

Kobayashi & Motoi propose path-planning methods that account for blind spots using ROS navigation and dynamic-window approaches — valuable for mobile-robot autonomy and safety. This is relevant only insofar as our stabilized ESP32 modules might be integrated into ROS-based robots as sensor nodes; the paper does not contribute to ESP32 power-stability or telemetry pipeline specifics. It remains orthogonal to our hardware-focused contributions.

Match Factor: 10%